

## Practice sheet: Inductors and Capacitors

This sheet is not marked for correctness, and AI tools are allowed. The device-free quiz is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Get both right; on the quiz each earns half the marks.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Conventions:** unless a figure shows otherwise, each element's current is referenced in the direction of its voltage drop (passive sign convention), so  $v = L \, di/dt$  and  $i = C \, dv/dt$ . An initial current or voltage that points against the reference is negative. Check every answer: put  $t = 0$  into your result, and differentiate it back.

**S1. An inductor in a dc circuit (two-tier).** A 2 H inductor in a dc circuit that has been on for a long time carries a constant 3 A. What is the voltage across it?

- |          |        |
|----------|--------|
| a) 6 V   | b) 0 V |
| c) 1.5 V | d) 3 V |

*Reason:*

- The voltage is  $v = Li$ .
- The voltage is  $v = L \, di/dt$ , and the current is not changing.
- The voltage is  $v = i/L$ .
- The voltage is the stored energy divided by the current,  $w/i = Li/2$ .

*Follow-up:* how much energy does the inductor store?

**S2. Voltage from current.** As in Nilsson Example 6.1. The current in a 0.5 H inductor is 0 for  $t < 0$  and  $i = 8te^{-10t}$  A for  $t \geq 0$ .

- When is the current largest, and what is it?
- Find  $v(t)$  for  $t > 0$ .
- When does the voltage change polarity?
- Does the voltage change instantly at  $t = 0$ ? Does the current?
- What is the largest energy the inductor stores?

**S3. Current from voltage.** As in Nilsson Example 6.2. A 2 H inductor carries  $i(0) = -1$  A. For  $t > 0$  the voltage across it is  $v = 12e^{-3t}$  V.

- Find  $i(t)$  for  $t \geq 0$ .
- When is the current zero?
- In what interval does the inductor deliver energy? Use the sign of  $p = vi$ .
- Find the stored energy at  $t = 0$  and as  $t \rightarrow \infty$ . Equal values: is that a contradiction?

**S4. A capacitor in a dc circuit (two-tier).** A  $5\ \mu\text{F}$  capacitor in a dc circuit that has been on for a long time has a constant  $12\ \text{V}$  across it. What current flows in it?

- a)  $60\ \mu\text{A}$
- b)  $30\ \mu\text{A}$
- c)  $0\ \text{A}$
- d)  $2.4 \times 10^6\ \text{A}$

*Reason:*

- i) The current is the stored energy divided by the voltage,  $w/v = Cv/2$ .
- ii) The current is  $i = Cv$ .
- iii) The current is  $i = v/C$ .
- iv) The current is  $i = C\ dv/dt$ , and the voltage is not changing.

*Follow-up:* how much energy does the capacitor store?

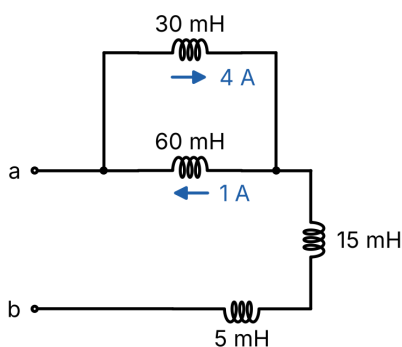
**S5. Current from voltage.** As in Nilsson Example 6.4. The voltage across a  $4\ \mu\text{F}$  capacitor is  $0$  for  $t < 0$  and  $v = 20(1 - e^{-500t})\ \text{V}$  for  $t \geq 0$ .

- a) Find  $i(t)$  for  $t > 0$ .
- b) Does the current change instantly at  $t = 0$ ? Does the voltage?
- c) Find the energy stored as  $t \rightarrow \infty$ .
- d) When is the power into the capacitor largest, and what is it?

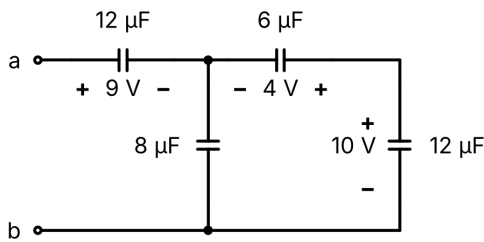
**S6. A current pulse.** As in Nilsson Example 6.5. A  $0.5\ \mu\text{F}$  capacitor has  $v(0) = 5\ \text{V}$ . A current of  $2\ \text{mA}$  flows into its  $+$  terminal from  $t = 0$  to  $t = 5\ \text{ms}$ , and zero afterwards.

- a) Find  $v(t)$  for  $0 \leq t \leq 5\ \text{ms}$ .
- b) What is  $v$  for  $t > 5\ \text{ms}$ ? Why does it not return to  $5\ \text{V}$ ?
- c) How much energy does the pulse deliver to the capacitor?
- d) *Predict first:* repeat (a)–(c) with a  $-2\ \text{mA}$  pulse. When does  $v$  pass through zero?

**S7. Equivalent inductance with initial currents.** As in Nilsson Example 6.6.



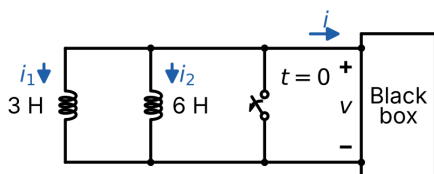
- a) Find  $L_{\text{eq}}$  between a and b.
- b) Find the initial current in  $L_{\text{eq}}$ , entering at a.
- c) The  $15\ \text{mH}$  and  $5\ \text{mH}$  carry the current found in (b). Find the energy stored in the four inductors and in  $L_{\text{eq}}$ . Where is the difference?

**S8. Equivalent capacitance with initial voltages.** As in Nilsson Example 6.7.

- Find  $C_{eq}$  between a and b.
- Find the initial voltage  $v_{ab}$  across  $C_{eq}$ .
- What is the initial voltage across the  $8 \mu F$ , + at the top? Find the energy stored in the four capacitors and in  $C_{eq}$ .

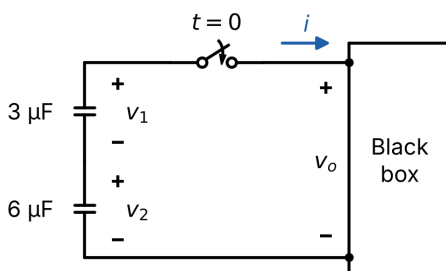
**S9. Parallel capacitors.** As in booklet Problem 6.9. A  $0.4 \mu F$  and a  $1.6 \mu F$  capacitor are in parallel. For  $t \geq 0$  the voltage across them is  $v = 50e^{-200t} + 30$  V.  $i_1$  is the current in the  $0.4 \mu F$  and  $i_2$  the current in the  $1.6 \mu F$ , each referenced into the + terminal.

- Find  $C_{eq}$ .
- Find  $i_1(t)$  and  $i_2(t)$ .
- Check that  $i_1 + i_2 = C_{eq} dv/dt$ . How do the two currents compare with the two capacitances?
- How much energy do the capacitors release between  $t = 0$  and  $t \rightarrow \infty$ ?

**S10. Inductors and a black box.** As in booklet Problem 6.8.

Just before the switch opens,  $i_1(0) = -9$  A and  $i_2(0) = 3$  A. For  $t > 0$ ,  $v = 36e^{-3t}$  V.

- Find  $L_{eq}$  and  $i(0^+)$ , the box current just after the switch opens.
- Find  $i(t)$ ,  $i_1(t)$  and  $i_2(t)$  for  $t \geq 0$ .
- Find the initial energy stored in the inductors.
- Find the energy delivered to the black box and the energy trapped in the inductors.

**S11. Capacitors and a black box.** As in booklet Problem 6.10.

At  $t = 0$ ,  $v_1(0) = 20$  V and  $v_2(0) = -8$  V. For  $t > 0$ ,  $i = 1.2e^{-50t}$  mA.

- Find  $C_{eq}$  and  $v_o(0)$ .
- Find  $v_1(t)$ ,  $v_2(t)$  and  $v_o(t)$  for  $t \geq 0$ .
- Find the initial energy stored in the capacitors.
- Find the energy delivered to the black box and the energy trapped in the capacitors.

**S12. Spot the error.** The worked solution below contains the kind of direction slip AI chat tools often make. A 4 H inductor has its current  $i$  referenced in the direction of its voltage drop  $v$ . At  $t = 0$  the current is 2 A, flowing against the reference arrow. For  $t > 0$ ,  $v = 24e^{-2t}$  V.

1.  $i(t) = (1/L) \int v \, dt + i(0)$ , integrating from 0 to  $t$ .
  2.  $i(0) = 2$  A.
  3. The integral of  $24e^{-2t}$  from 0 to  $t$  is  $12(1 - e^{-2t})$ .
  4.  $i(t) = (12/4)(1 - e^{-2t}) + 2 = 5 - 3e^{-2t}$  A.
  5. Check:  $i(0) = 2$  A and  $4 \, di/dt = 24e^{-2t}$  V, so the answer is right.
- a) Which is the first wrong line?
  - b) Correct it and find  $i(t)$ .
  - c) Line 5 calls itself a check. Why did it not catch the error?
  - d) When does the current reverse direction?

**S13. Spot the error in an energy balance.** A 2 H and an 8 H inductor are in parallel, with  $i_1(0) = 6$  A and  $i_2(0) = -1$  A, both referenced downward; until  $t = 0$  a switch across their terminals carries the total current. At  $t = 0$  the switch opens and the current flows into a black box, which draws current until the total current is zero.

1.  $L_{eq} = (2)(8)/(2 + 8) = 1.6$  H.
  2. The initial stored energy is  $(1/2)(2)(6)^2 + (1/2)(8)(1)^2 = 36 + 4 = 40$  J.
  3. The box draws current until the inductors are empty, so it receives all the stored energy.
  4. The energy delivered to the box is 40 J.
- a) Which is the first wrong line?
  - b) Find the energy delivered to the box. (Use  $L_{eq}$  and the total initial current.)
  - c) Find  $i_1$  and  $i_2$  after a long time, and the energy trapped.

**S14. A supercapacitor bank.** A supercapacitor cell is rated 3000 F at 2.7 V.

- a) How much energy does one cell store at its rated voltage?
- b) Six cells are connected in series for a 16.2 V bank. Find  $C_{eq}$  and the energy the bank stores at 16.2 V. Compare with six times the energy of one cell.
- c) For how long could the bank supply 200 W if all of that energy could be used?
- d) *Predict first:* two such banks are put in parallel. What happens to  $C_{eq}$  and to the stored energy?

**S15. Capacitive touch screens.** As in Nilsson's Practical Perspective for this chapter. Each electrode of a touch screen has a parasitic capacitance  $C_p = 20$  pF to ground. A finger adds  $C_t = 5$  pF at the point of touch.

- a) What capacitance does the controller see at the touched electrode? By what percentage did it change?
- b) A self-capacitance screen is touched at (X1, Y2) and (X3, Y0) at the same time. Which four grid points look touched? Which two are ghosts?
- c) Why does a mutual-capacitance screen not produce ghost points?
- d) The controller charges an electrode with a constant 1  $\mu$ A. How long does the voltage take to reach 1 V when untouched, and when touched?

**S16. Check an answer without solving again.** A 0.25 H inductor carries  $i(0) = 3$  A. For  $t > 0$ ,  $v = 10e^{-20t}$  V. Two classmates report  $i(t)$  for  $t \geq 0$ . Student 1:  $i = 5 - 2e^{-20t}$  A. Student 2:  $i = 1 + 2e^{-20t}$  A. Use  $i(0)$  and  $v = L \, di/dt$  to decide which student is right.